CTCH 605
Introduction to Cybersecurity

Security Assessment Report

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Version 1.0

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1. Executive Summary

The owner of FICBANK contracted me as a cybersecurity professional to perform a vulnerability assessment and security monitoring review of the credit union IT infrastructure. The objective of this assessment was to identify open ports, detect system vulnerabilities, and analyze FICBANK's ability to collect and query security monitoring data.During the assessment, I utilized Nmap and Nessus to scan the target host, windows VM *10.15.36.103*.

Nmap detected nine open ports and services which included:
22/tcp (SSH), 135/tcp (MSRPC), 139/tcp (NETBIOS-SSN), 445/tcp (MICROSOFT-DS), 1947/tcp (HTTP), 2179/tcp (VMRDP), 3389/tcp (MS-WBT- SERVER), 5357/tcp (HTTP), and 5985/tcp (HTTP).

In addition, Nessus identified a total of 642 vulnerabilities, categorized as follows: 115 critical, 189 high, 82 medium, 256 Informational. The scan generated 27 remediation recommendations to address the identified threats.

The assessment confirms that FICBANK's system has multiple open port, unpatched softwares, and minimize security monitoring. As a result, the system is highly susceptible to attacks if security measures are not implemented immediately.

2. Assessment Scope

The technical infrastructure assessment was to analyze FICBANKS internal computer system, which included one target host: windows VM *10.15.36.10.* The objective of this assessment was to identify open ports, detect system vulnerabilities and evaluate the credit union's ability to monitor its security systems.

The vulnerability scanning tools used during the assessment were Nmap and Nessus. Each tool played a significant role in identifying different components of system vulnerabilities. Nmap was used to detect open ports and active service within the system. While Nessus was used to identify host vulnerabilities and categorize each threat from highest to lowest risk level.
Due to the credit union's limited resources, I expected to find multiple internal and external vulnerabilities. Most of the internal vulnerabilities consisted of outdated software, weak passwords, and unsecured internal servers. The external vulnerabilities included open ports, misconfigured VPNs, and phishing attacks.

Overall, the assessment confirmed nine open ports, with two high risk ports identified as 3389/tcp (MS-WBT- SERVER) and 445/tcp (MICROSOFT-DS). The Nessus scan reported a total of 642 vulnerabilities, including 115 critical and 189 high risk vulnerabilities. In addition, no active security monitoring system was detected within the credit union's IT infrastructure.

3. Assessment Methodology

As a cybersecurity professional contracted by FICBANK, My objective was to test the credit union's network for vulnerabilities and evaluate the security monitoring system. Below are steps performed to identify open ports and discover vulnerabilities:

Step 1: Identify the target host

First, identify the target host by using Windows command prompt IPconfig to find the system IP address, 10.15.36.103. This IP address will be used in the Nmap and Nessus vulnerability assessment.

Step 2: Update and install Nmap

Next, I opened the Kali Linux operating system to access the Nmap vulnerability scan tool. To access the Nmap on the Kali Linux system I selected application, system tools then Nmap. Before launching the Nmap scanner, it is important to confirm the scan is installed and updated by typing commands: Sudo apt update and sudo apt install nmap. This process ensures Nmap is papered to scan for open ports within the network.

Step 3: Perform Nmap Scan to identify open ports

Nmap is ready to scan for the target host Windows VM 10.15.36.103 for open ports. These open ports are referred to as "digital doors" which allow networks the ability to connect with each other. However, having too many open ports can increase system vulnerability if a cyberattacker gains access to the network. To begin the scan, I opened the terminal in Nmap and type the command, sudo nmap - sV 10.15.36.103. This command identifies nine open ports and each active services. Two of the open ports were considered high risk, 445/tcp and 3389/tcp. These open ports include file sharing and remote desktop which is a widely common target for a cyberattack. After Nmap identified the open ports and active service running on the system, I saved the results using command car scan_results.txt.

Step 4:Perform Nessus scan to discover vulnerabilities

After identifying the open ports in Nmap, I performed a vulnerability scan using the Nessus tool. This scan helps to discover vulnerabilities within the target system, such as outdated software and missing patches.

The vulnerabilities are categorized by critical to informational threats. To begin the scan, I opened the Kali Linux operating system and launched the Firefox web browser. In the address bar, I entered https://localhost:8834

This opened the Nessus web interface. I signed in using my credentials then selected "New Scan" followed by the type of scan, Basic Network Scan. I entered the target host, IP address 10.15.36.103 and configured the scan settings, including the scan name, target details, scan schedule, and credentials. Once the scan was complete, Nessus discovered a total of 643 vulnerabilities categorized as 115 critical, 189 high, 82 medium, and 256 informational. The results confirmed that the system contained multiple outdated and unpatched software components. After reviewing the results, I saved the report for documentation.

Step 5: Review and Compare Results

Now that the Nmap and Nessus scans are complete, it was important to review and compare the results. By comparing the results I can interpret the overall vulnerability with the credit union's IT infrastructure.

The Nmap scan identified nine open ports, including two high risk open ports, 445/tcp and 3389/tcp. These open ports are commonly targeted by attackers to gain unauthorized access or to spread malware.

The Nessus scan provided a more comprehensive look into the system's vulnerabilities. The assessment reported a total of 642 vulnerabilities categorized by severity as follows:

- 115 Critical
- 189 High
- 82 Medium
- 256 Informational

These findings indicate that the target system contains several outdated or unpatched software components while also lacking proper security controls. The high number of critical and high severity vulnerabilities propose that an immediate remediation is necessary to reduce the system's risk exposure.

By comparing both scans, it became apparent that several of the open ports discovered in the Nmap are directly linked to the vulnerabilities identified by Nessus. For example, services running on ports 445 and 3389 were associated with multiple critical vulnerabilities. This confirms that they are major points of concern for the system.

4. Detailed Findings

Step 1: Performed an Nmap Scan to identify Open Ports

- **Action:** I opened the Kali Linux operating system to access the Nmap vulnerability scan tool to target host Windows VM 10.15.36.103 typing command *sudo nmap - sv 10.15.36.103* to identify open ports and active services.

- **Results:** The Nmap scan identified nine open ports and active services running on the system. Each open port represent potential vulnerability within the system, allowing cyberattacker access within the internal systems. If left unpatched, these vulnerabilities could result in data loss or system damage within the IT infrastructure. There are two ports, 445/tcp and 3389/tcp, require immediate remediation, as they are considered high risk due to the level of accessibility they provide though file-sharing and remote desktop services.

Below is a list out the open ports and active service running on the network:

Port	Service	Description
22/tcp	SSH	Secure remote access software that connect to another computer or file transfer
135/tcp	MSRPC	Window system service that Remotely connect computers together to share task
139/tcp	NETBIOS-SSN	Window network sharing service that allows file and printer sharing between computer
445/tcp	Microsoft-DS	Windows file sharing to communicate on the same network
1947/tcp	HTTP	Software licensing service that manages the use of authorized software
2179/tcp	VMRDP	Virtual machine remote desktop service that is used to allow remote connection to a compute
3389/tcp	MS-WBT-Server	Remote Desktop software that is used to control another computer screen
5357/tcp	HTTP	Device discovery service that help find printer or scanner devices on the network
5985/tcp	HTTP	Used as a windows remote software that allows admins to manage computer remotely

Nmap Assessment Screenshots:

Image 1: Displays the update of Kaili Linux using the commend *sudo apt update*

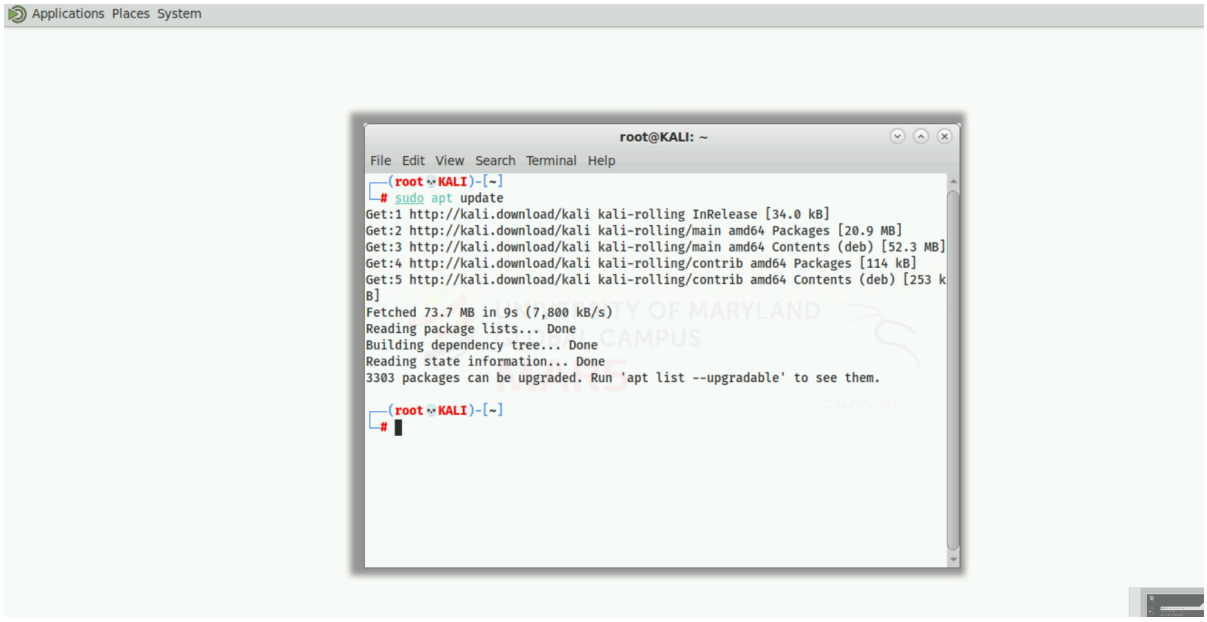


Image 2: Displays the installation of of Nmap using the commend *sudo install Nmap*

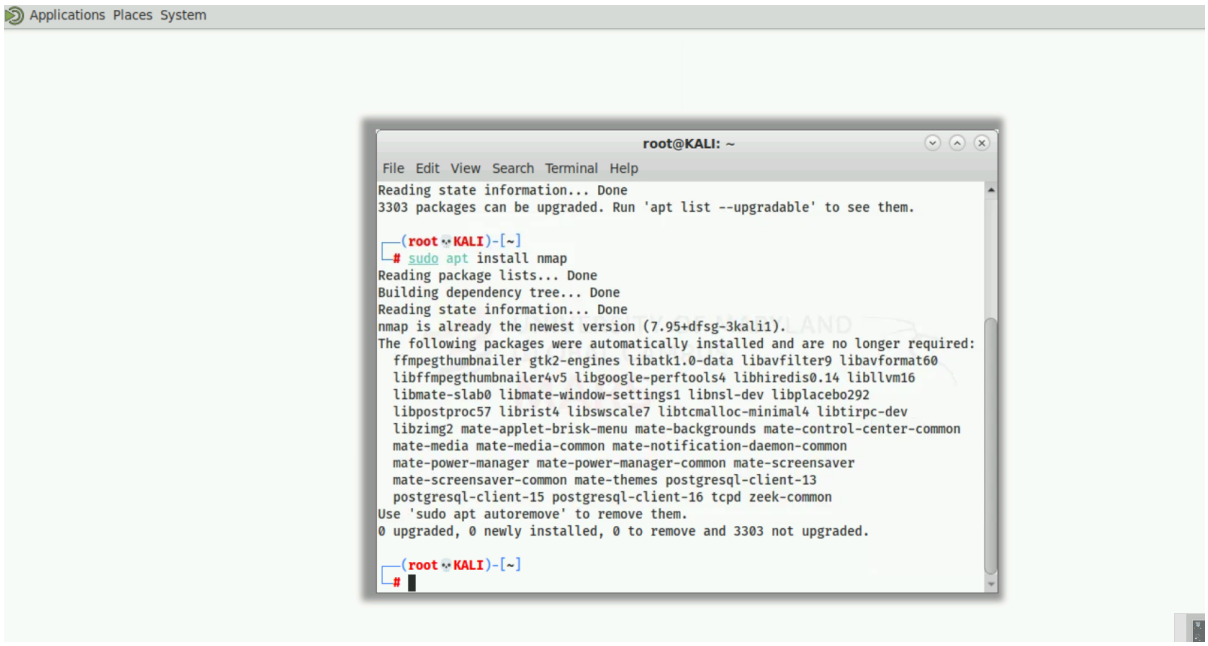


Image 3: Displays the execution of Nmap scan on the target host 10.15.36.103 using the commend *sudo nmap -sV 10.15.36.103* to identify open ports and active service running on system. The result shows nine open ports.

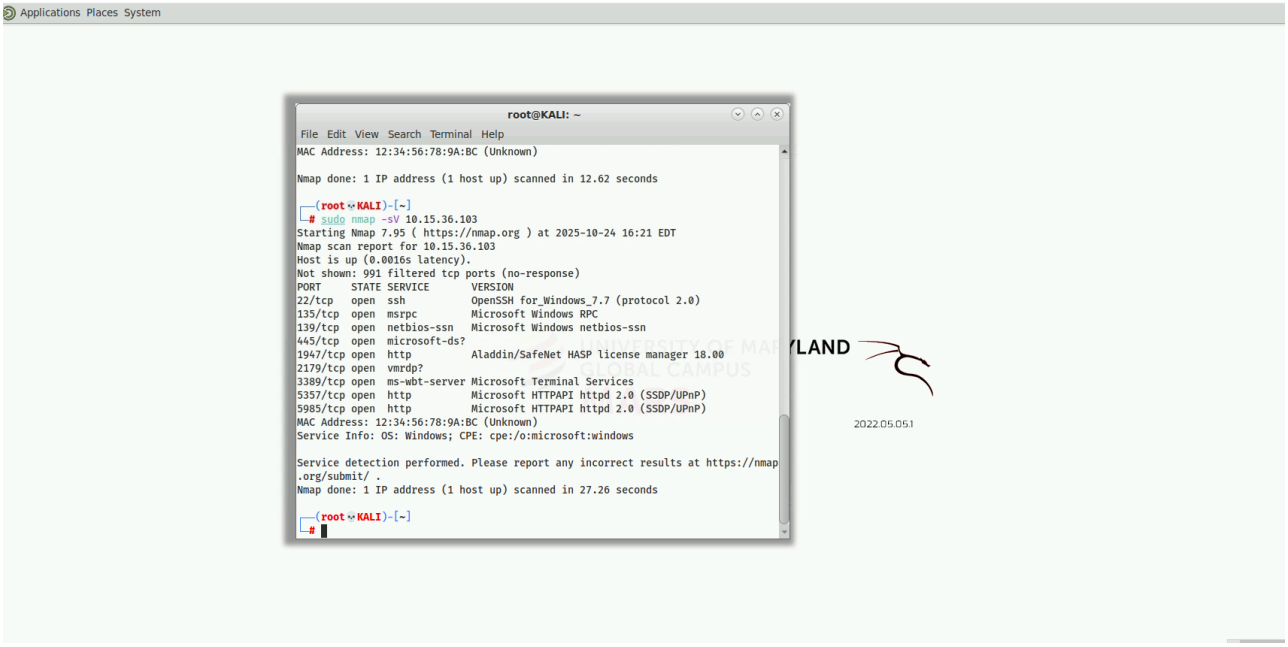
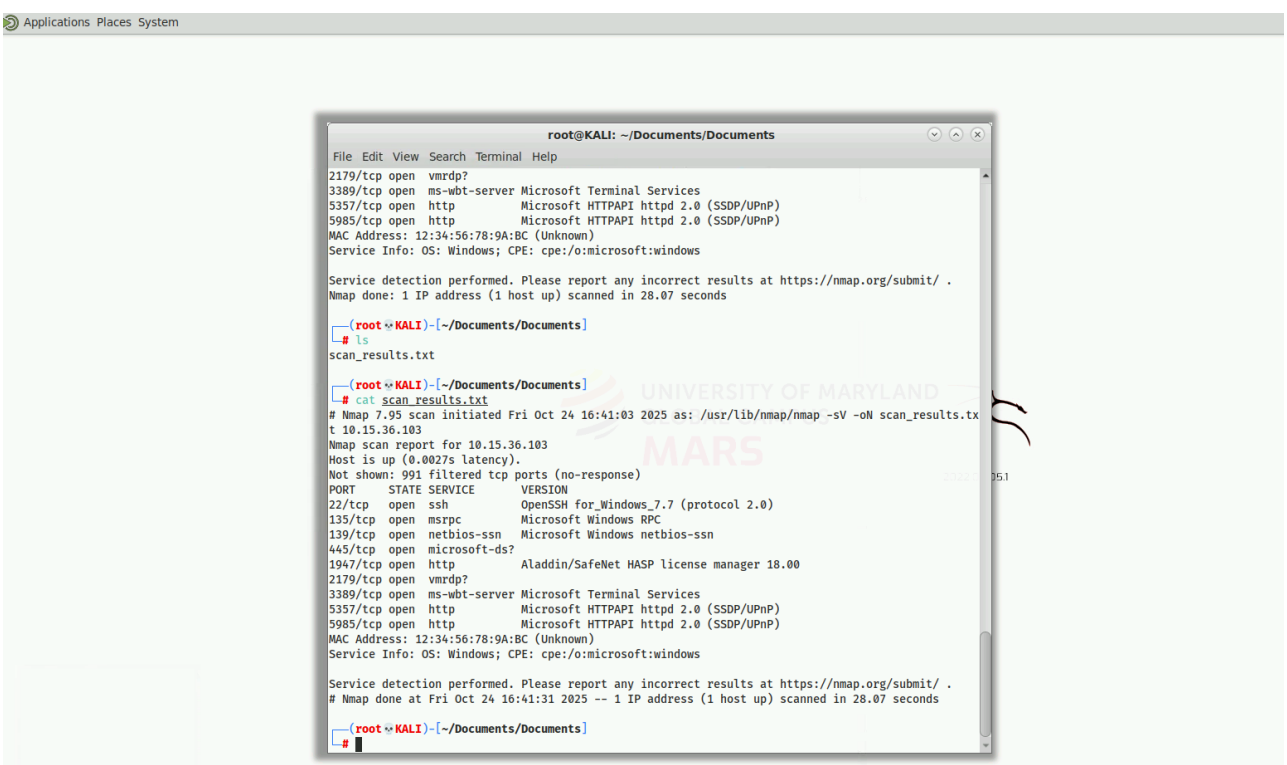


Image 4: Displays the saving and viewing of Nmap scan results of the target host 10.15.36.103 using the commend *cat scan_results.txt.* This image verifies that scan results have been documented.



Step 2: Performed an Nessus Scan to discover vulnerabilities within the network

- **Action:** After open ports were identified using Nmap, I performed a vulnerability scan using the Nessus vulnerability tool on the target host Windows VM 10.15.36.103. I ran a Basic Network Scan, on the FICBANK's IT infursturture to detect vulnerabilities caused by outdated softwares, unsecured configurations, and missing updates.

- **Result:** The Nessus vulnerability Scan discovered a total of 642 vulnerabilities on target host Windows VM 10.15.36.103. Each vulnerabilities represents a potential threat that could allow cyber attacker to exploit weaknesses within the credit union's IT infrastructure, if not properly addressed.

Below are the vulnerabilities categorized by severity levels:

- 115 Critical
- 189 High
- 83 Medium
- 256 Informational

Most of these vulnerabilities were linked to outdated or unpatched software. These softwares includes Mozilla Firefox, Oracle Save, Visual Studio, Wireshark, and MySQL Server. These unpatched software increases the risk of data breaches, unauthorized remote access or system compromise within the FICBANK's IT infrastructure. Overall, immediate remediation is recommended for all the Critical and High vulnerabilities to prevent cyber threats and strengthen overall system.

- **Nessus Assessment Screenshot:**

Image 1: Displays the configuring of target host to the Nessus Scan tool

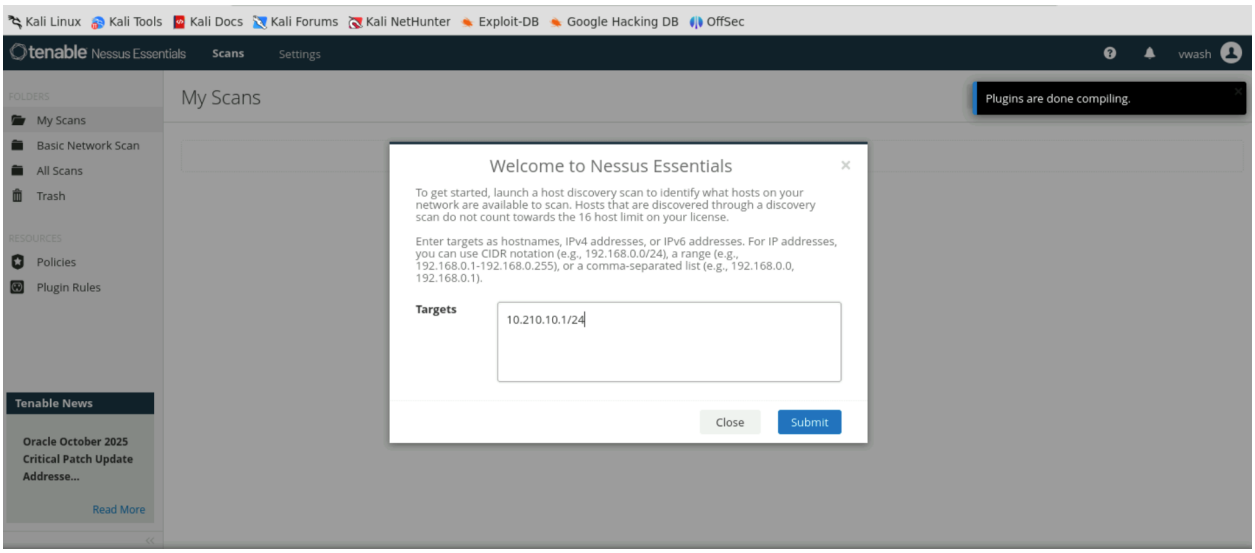


Image 2: Displays the Host Discovery Scan Results. The results identified two active hosts *10.15.36.100* and *10.15.36.103*. This image confirms that these systems are ready for vulnerability scanning.

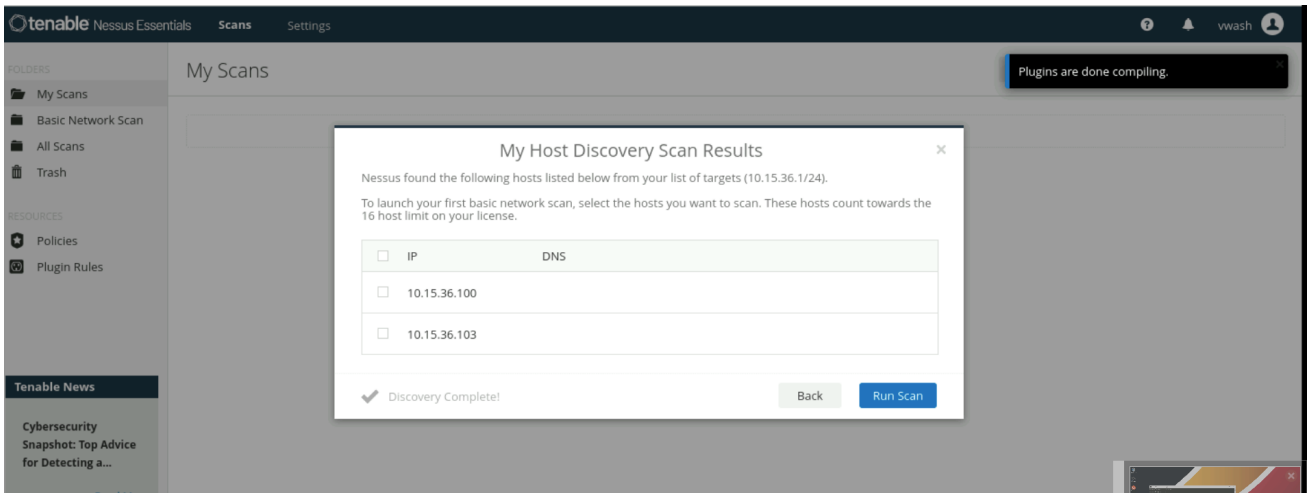


Image 3: Displays the Basic Network Scan Result for target host ***10.15.36.103***. The results identified two active hosts *10.15.36.100* and *10.15.36.103*. This image shows the results of multiple vulnerabilities categorized by Critical, high, medium, low, and informational. The right panel shows scan details including policy, status, severity base, scanner, and elapsed time.

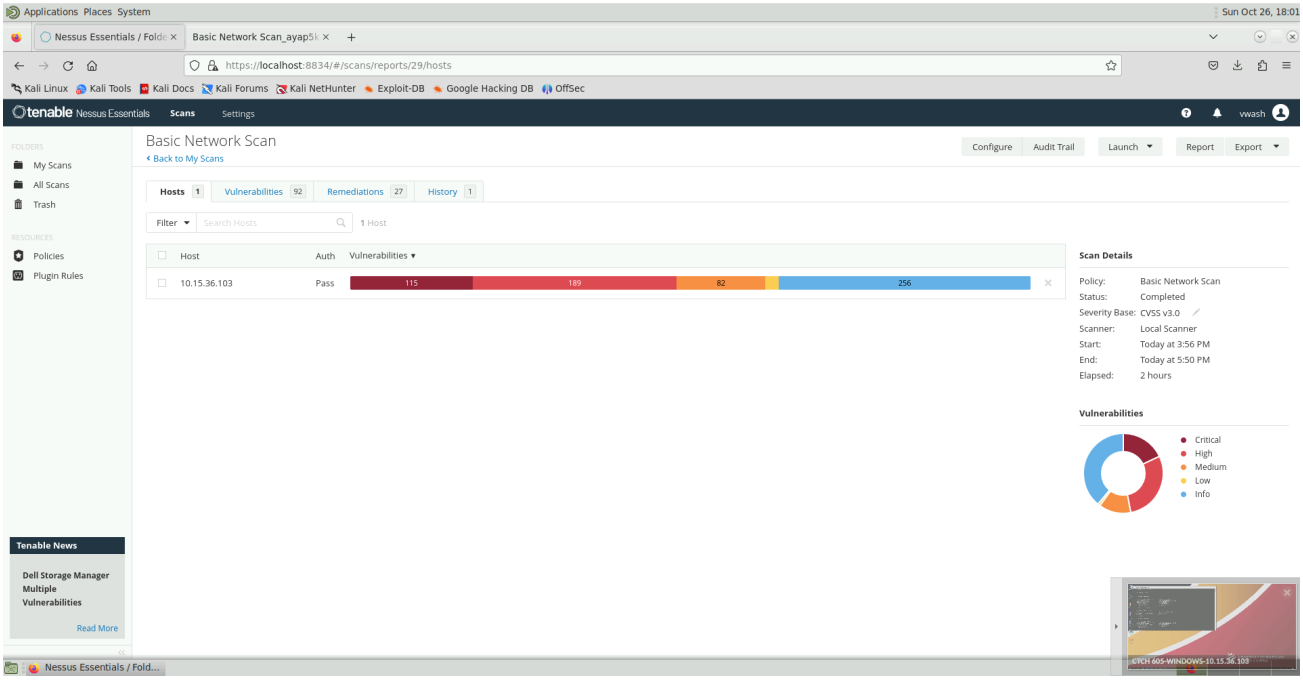


Image 4: Displays the detailed vulnerability results for target host ***10.15.36.103***. The results listed all the detected vulnerabilities categorized by software name and severity level. The unpatched softwares include Mozilla Firefox, Oracle JRE, Microsoft Visual Studio, MySQL.

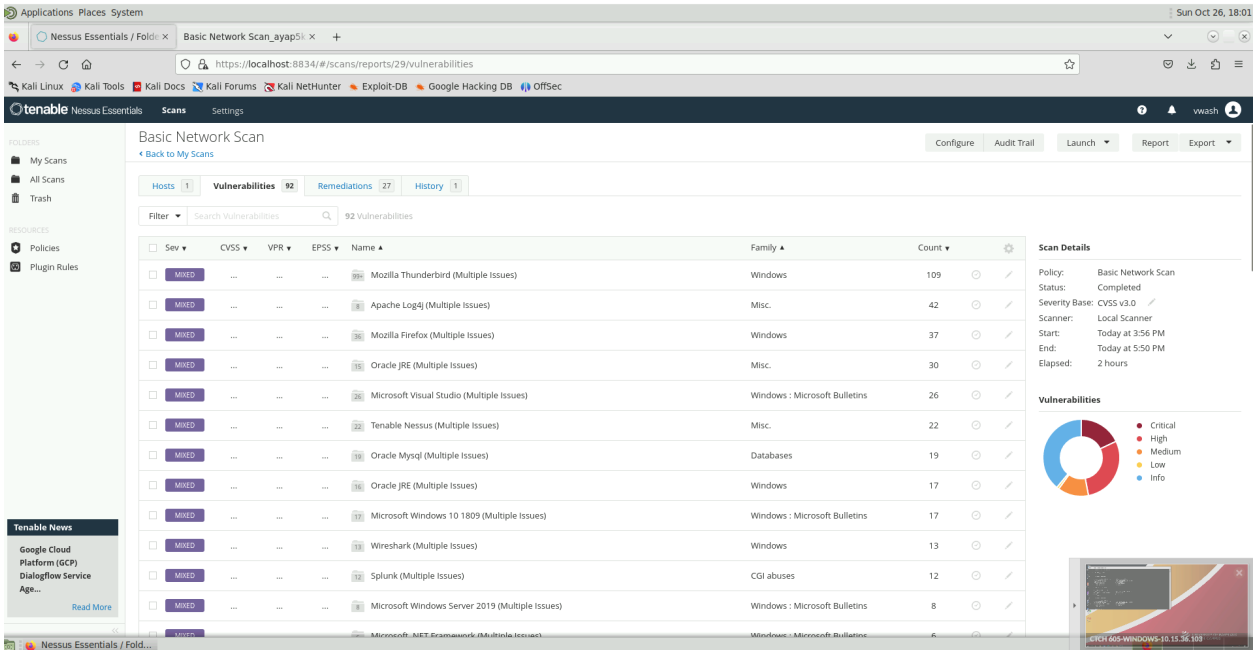
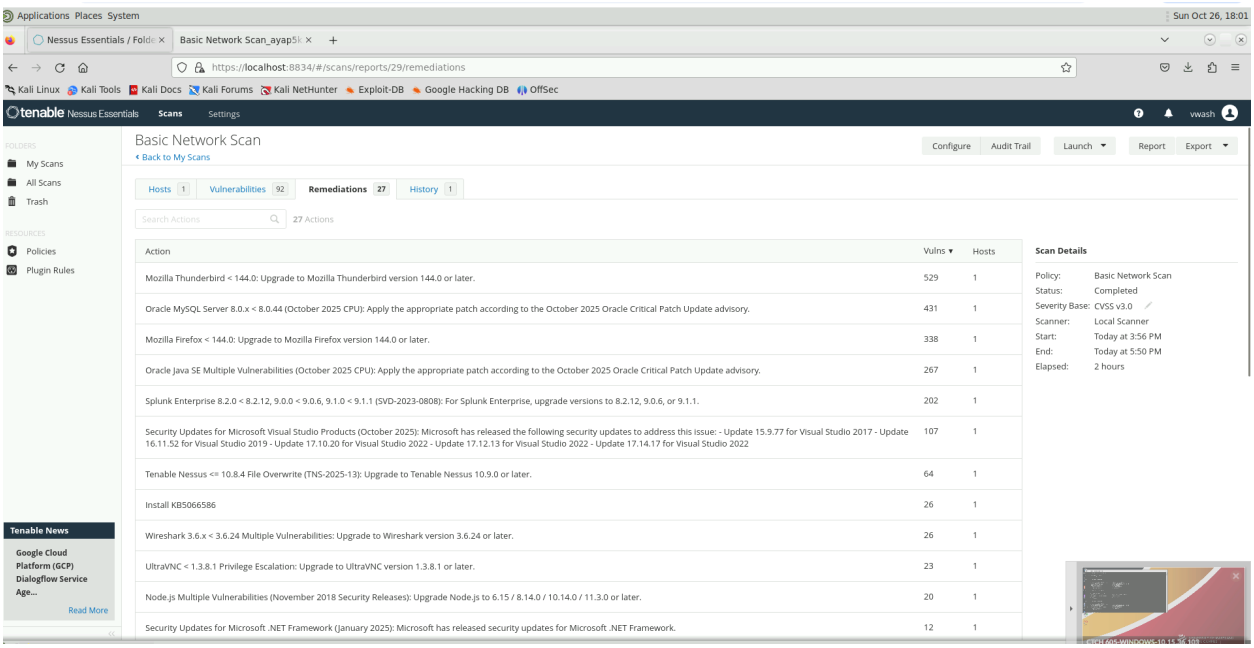


Image 5: Displays the Remediation results for the target host ***10.15.36.103***. The results list all recommended actions to address the discovered vulnerabilities.



5. Conclusion

1. Assessment Summary:

As a cybersecurity professional, my objective was to identify open ports, detect system vulnerabilities, and elevate FICBACK's ability to collect and monitor security data. The security assessment confirmed that there are multiple vulnerabilities within the credit union's IT infrastructure. The vulnerability scanning tools used during this process were Nmap and Nessus.

The Nmap scan identified nine open ports and active service running on the system. Out of the nine, two open ports were considered high risk, 445/tcp Microft-ds and 3389/tcp MS-WBT-Server. These open ports are commonly used for file sharing and remote desktop access, which can allow cyber attackers to gain unauthorized access or spread malware's, if left unsecured.

The Nessus scanner was used to detect vulnerabilities within the system. The scan detected a total of 642 vulnerabilities, including 115 critical, 189 high risk, 83 medium, and 256 informational. These results confirmed that FICBANK's system contains outdated softwares, unsecured configurations, and missing security updates, making the credit union highly susceptible to potential cyberattack.

2. Recommend Remediation Techniques:

Due to the number of vulnerabilities and open ports identified within FICBANK's system, Nessus generated a remediation report recommending multiple actions to reduce the risk of cyberattacks. The report recommends updating outdated software programs, such as Mozilla Firefox, Oracle Java, MySQL Server, Microsoft Visual Studio and Wireshark. In addition to these software updates, it important to apply the latest Windows security patches to ensure the system is protected.

Furthermore, closing unnecessary open ports and restricting external remote desktop access and limiting file sharing capabilities will improve the security of system. Implementing the recommendations will help prevent unauthorized access, minimize malware exposure and strengthen the security of FICBANK's IT infrastructure.

3. Remediation and Future Impact improvement

I highly recommend FICBANK to implement these remediation steps to improve its security system. These actions will significantly reduce the number of critical and high risk vulnerabilities found in FICBANK's system. Regularly applying software updates, closing unused ports, and restricting remote access will help prevent attackers from taking advantage of weaknesses within the network. This will also protect sensitive customer data and prevent service disruptions. Taking these actions will improve the overall security of FICBANK's IT infrastructure.

I recommend the following improvements for maintaining and strengthening system security:

- Continuous Monitoring: Set up a system within the credit union that collects and monitors security logs in real time.
- Patch Management: Regularly schedule software updates and systems to fix vulnerabilities quickly.
- Routine Scanning: Use Nmap and Nessus to detect new vulnerabilities early and perform scans regularly.
- Access Control: Utilize multi factor authentication (MFA) and strong passwords for user accounts
- Employee Training: Implement and enforce cybersecurity awareness training for all employees to avoid unsafe practices.

FICBANK can maintain stringer protection against cyber threats, improve the IT infrastructure, and ensure that customer data is safe from loss or unauthorized access by following the above recommendations.

4. Reflection Question and Response

How did the FICBANK vulnerability assessment help me understand the importance of identifying, analyzing and remediation techniques to assess system vulnerabilities?

This hands-on assessment provided me with practical, real life experience in the field of cybersecurity. There was a large amount of new information to learn and understand within a short time frame. However, I was able to apply the skills and concepts covered in Chapter 1 and 2 to a real system, which helped me connect what I learned in class to actual cybersecurity work.

At the beginning of the project, I underestimated the number of vulnerabilities that could exist within a IT system. I expected to find a few outdated software programs or weak passwords but the assessment results showed that system cybersecurity can be much more complex. The experience helped me understand that cyberattacks can occur in many different ways and that identifying and resolving vulnerabilities quickly is essential to maintaining a secure network.

Using Nmap and Nessus during the assessment helped me understand how different cybersecurity tools serve specific purposes. Nmap was used to identify open ports, which may appear harmless but can create serious risks if they are not properly secured. Nesssus, on the other hand, provided detailed information about each vulnerability, categorizing them by severity and recommended remediation steps. These tools helped me learn how to analyze, prioritize and address risks within the system.

Overall, this assessment taught me the importance of keeping software updated, monitoring systems regularly, and maintaining cybersecurity awareness across all levels of an organization. It challenged my ability to complete real world cybersecurity tasks and strengthened my understanding of how technical knowledge is applied in practice. I would recommend including more hands on projects like this in cybersecurity programs because they provide realistic practice, as they also help build confidence in performing security assessments.